HW1.6

January 23, 2026

Read about automatic differentation in "Machine Learning Defined" appendix B. You can find that online here https://jermwatt.github.io/machine_learning_refined/notes/3_First_order_methods/3_5_Automat
To make the module autograd available we use the following commands:

[2]:
```
# import statement for autograd wrapped numpy
import autograd.numpy as np
# import statment for gradient calculator
from autograd import grad
```

Then we can use the following generic code for doing the gradient descent:

[4]:
```
# import automatic differentiator to compute gradient module
from autograd import grad

# gradient descent function - inputs: g (input function), alpha (steplength␣
 ↪parameter), max_its (maximum number of iterations), w (initialization)
def gradient_descent(g,alpha,max_its,w):
    # compute gradient module using autograd
    gradient = grad(g)

    # run the gradient descent loop
    weight_history = [w]           # container for weight history
    cost_history = [g(w)]          # container for corresponding cost function␣
 ↪history
    for k in range(max_its):
        # evaluate the gradient, store current weights and cost function value
        grad_eval = gradient(w)

        # take gradient descent step
        w = w - alpha*grad_eval

        # record weight and cost
        weight_history.append(w)
        cost_history.append(g(w))
    return cost_history
```

We now want to find the minimum of

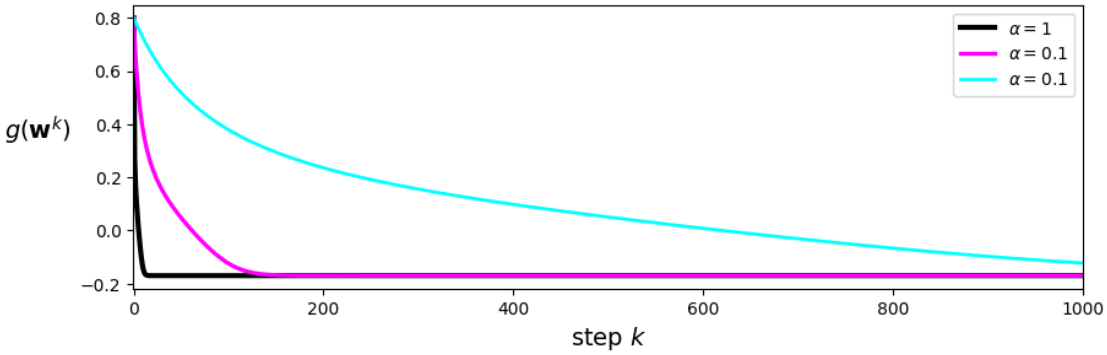
$$g(w) = \frac{1}{50}\left(w^4 + w^2 + 10w\right)$$

using the gradient descent method described above. Make three separate runs using step lengths $\alpha = 1$, 0.1, and 0.01, starting from $w^0 = 2$ with 1000 iterations.

Plot the resulting cost histories. Which step length works best for this particular function and initial point?

[8]:
```
# defining g
g = lambda w: (w**4 + w**2 + 10*w) / 50

# parameters
alphas= [1, 0.1, 0.01]
w0 = 2.0
max_its = 1000

# cost histories
cost_histories_1 = gradient_descent(g,alphas[0],max_its,w0)
cost_histories_2 = gradient_descent(g,alphas[1],max_its,w0)
cost_histories_3 = gradient_descent(g,alphas[2],max_its,w0)
```

[11]:
```
from MLRPlotting import static_plotter

# plotting the cost histories
from MLRPlotting import static_plotter

# import custom plotting tools
static_plotter = static_plotter.Visualizer()
static_plotter.
 ↪plot_cost_histories([cost_histories_1,cost_histories_2,cost_histories_3],start=0,points=Fal
 ↪= 1$',r'$\alpha = 0.1$',r'$\alpha = 0.01$'])
```



[13]:
```
# minimum found for each alpha
print("Minimum found with alpha = 1: w* = ", np.round(np.
  ↪array(cost_histories_1[-1]),4))
print("Minimum found with alpha = 0.1: w* = ", np.round(np.
  ↪array(cost_histories_2[-1]),4))
print("Minimum found with alpha = 0.01: w* = ", np.round(np.
  ↪array(cost_histories_3[-1]),4))
```

```
Minimum found with alpha = 1: w* =  -0.17
Minimum found with alpha = 0.1: w* =  -0.17
Minimum found with alpha = 0.01: w* =  -0.1225
```

both alpha = 1 and alpha = 0.1 reaches the minimum but alpha = 1 decreases faster.